

Homework Set 4

Problem 1: We would like to evaluate the result of the first-visit Monte Carlo algorithm used for policy evaluation in the episodic, discounted setting. Recall that this algorithm estimates the value function under policy π in any state s , i.e.,

$$V^\pi(s) = \mathbb{E}_{\substack{A_t \sim \pi(\cdot|S_t) \\ S_{t+1} \sim P(\cdot|S_t, A_t)}} \left[\sum_{t=0}^T \gamma^t R(S_t, A_t) | S_0 = s \right]$$

by the empirical mean of a set of sample returns. Let $\mathcal{T}(s) = \{\tau_1^s, \tau_2^s, \dots, \tau_{N^s}^s\}$ be the sample trajectories starting from s and ending in a terminal state obtained from different episodes of the algorithm. Let $\mathcal{G}(s) = \{G_1^s, G_2^s, \dots, G_{N^s}^s\}$ be the sample returns corresponding to $\mathcal{T}(s)$, i.e.,

$$G_i^s = \sum_{t=0}^{|\tau_i^s|} \gamma^t R(S_{t,i}^s, A_{t,i}^s),$$

where $S_{t,i}^s$ and $A_{t,i}^s$ are the state and action observed at time t in trajectory τ_i^s , respectively. The Monte Carlo algorithm estimates $V^\pi(s)$ as follows:

$$\hat{V}^\pi(s) = \frac{1}{N^s} \sum_{i=1}^{N^s} G_i^s = \frac{1}{N^s} \sum_{\tau_i^s \in \mathcal{T}(s)} \sum_{t=0}^{|\tau_i^s|} \gamma^t R(S_{t,i}^s, A_{t,i}^s).$$

1. Assume that the reward function is bounded, $|R(s, a)| \leq R_{max}$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$. Find an upper bound and lower bound on G_i^s , i.e., find α and β such that

$$\alpha \leq G_i^s \leq \beta.$$

2. Let $E(s) = \sum_{i=1}^{N^s} G_i^s$ be the sum of all sample returns for state s . Recall that the expected value of each sample return is the true value function, i.e., $\mathbb{E}_{\substack{A_t \sim \pi(\cdot|S_t) \\ S_{t+1} \sim P(\cdot|S_t, A_t)}} [G_i^s] = V^\pi(s)$.

Derive and express $\mathbb{E}_{\substack{A_t \sim \pi(\cdot|S_t) \\ S_{t+1} \sim P(\cdot|S_t, A_t)}} [E(s)]$ in terms of $V^\pi(s)$.

3. Apply Hoeffding's inequality to bound the probability that $E(s)$ deviates from its expected value obtained in the previous part, i.e., bound $\mathbb{P}(|E(s) - \mathbb{E}[E(s)]| \geq \epsilon)$ (the subscript for the expectation operator is omitted for simplifying the notation) for any $\epsilon > 0$. Notice that the samples G_i^s are independent random variables.
4. Now, considering $\hat{V}^\pi(s) = \frac{1}{N^s} E(s)$, bound the probability that $\hat{V}^\pi(s)$ deviates from $V^\pi(s)$, i.e., bound $\mathbb{P}(|\hat{V}^\pi(s) - V^\pi(s)| \geq \epsilon')$ for any $\epsilon' > 0$.
5. Suppose that we have collected N^s sample trajectories for each state $s \in \mathcal{S}$, and let $N = \max_{s \in \mathcal{S}} N^s$. Let $V^\pi \in \mathbb{R}^{|\mathcal{S}|}$ and $\hat{V}^\pi \in \mathbb{R}^{|\mathcal{S}|}$ be the true and estimated value function, respectively, for all states $s \in \mathcal{S}$ represented as column vectors. Derive an upper bound on the probability that $\hat{V}^\pi$ deviates from V^π , i.e., bound $\mathbb{P}(\|\hat{V}^\pi - V^\pi\|_\infty \geq \epsilon')$ for any $\epsilon' > 0$. (The bound should be written in terms of N and not N^s .)

[Hint: You may find the union bound (Boole's inequality),

$$\mathbb{P}\left(\bigcup_{i=1}^n B_i\right) \leq \sum_{i=1}^n \mathbb{P}(B_i) \quad \text{for any set of events } B_1, B_2, \dots, B_n,$$

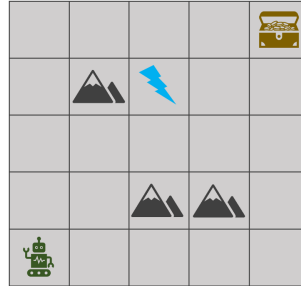
helpful for this part.]

Problem 2: Consider a Markov decision process (MDP) in the episodic setting with state space $\mathcal{S} = \{b, c, d\}$, action space $\mathcal{A} = \{x, y\}$, an unknown transition function, an unknown reward function, and discount factor γ . The episodes terminate once the agent reaches state $s = c$. We want to use Monte Carlo evaluation to estimate the state-action value function at state $s = b$ and $a = x$ for a policy π , i.e., $Q^\pi(b, x)$. We have collected three sample trajectories from the MDP based on this policy, as listed below.

	$t = 0$	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$
τ_1	$s_0 = d$	$s_1 = b$	$s_2 = d$	$s_3 = c$			
	$a_0 = x$	$a_1 = x$	$a_2 = y$				
	$r_0 = 1.5$	$r_1 = -1$	$r_2 = 2$				
τ_2	$s_0 = b$	$s_1 = d$	$s_2 = b$	$s_3 = b$	$s_4 = d$	$s_5 = d$	$s_6 = c$
	$a_0 = x$	$a_1 = y$	$a_2 = y$	$a_3 = x$	$a_4 = x$	$a_5 = x$	
	$r_0 = -1$	$r_1 = 2$	$r_2 = 0$	$r_3 = -1$	$r_4 = 1.5$	$r_5 = 1.5$	
τ_3	$s_0 = b$	$s_1 = d$	$s_2 = d$	$s_3 = d$	$s_4 = b$	$s_5 = c$	
	$a_0 = y$	$a_1 = x$	$a_2 = x$	$a_3 = y$	$a_4 = y$		
	$r_0 = 0$	$r_1 = 1.5$	$r_2 = 1.5$	$r_3 = 2$	$r_4 = 0$		

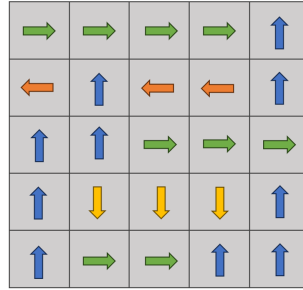
Apply an every-visit Monte Carlo method to estimate $Q^\pi(b, x)$ based on the collected sample trajectories. The answer will be in terms of γ .

Problem 3: We want to evaluate a policy for the same example in Problem 3 of Homework 3 in a model-free setting, i.e., when the environment model (the transition function and the reward function) is unknown to the agent. Consider an MDP over the grid-world illustrated below, where the goal is to take the agent to the treasure chest while avoiding the lightning.



- The state space contains the cells of the grid-world.
- The action space is {up, down, left, right}.
- The agent can only move to its adjacent cells, i.e., the cells that are above, below, to the left, or to the right of its current cell. If the agent is not at the boundary cells, each action will take the agent to the expected cell with probability 0.85 and to one of the remaining three cells, each with probability 0.05. If the agent is at the boundary, it will remain at its current cell with the sum of probabilities that would have taken it outside the grid-world. The cell with a mountain cannot be accessed, i.e., if adjacent to the mountain, the agent will remain at its current cell with the probability that would have taken it to the mountain. All actions in the cell with the lightning bolt, the one with the treasure chest, and the ones with a mountain will keep the agent in its current cell.
- The agent receives a reward of 0 in every cell for all actions except two cells. If at the cell with a lightning bolt, it will receive a reward of -1 for all actions, and if at the cell with the treasure chest, it will receive a reward of $+1$ for all actions.
- The discount factor is 0.95.
- Model the setting as an episodic one, where each episode ends once the agent is at one of the terminal states, i.e., the cell with the lightning bolt, the one with the treasure chest, and the ones with a mountain. Once the agent reaches those terminal states, consider that the agent takes one action in those states, receives its reward value, and then, the episode terminates. Once an episode ends, the state resets to (agent respawns at) an initial state in the next episode.

Evaluate the deterministic, stationary policy shown in the figure below, where each arrow represents the prescribed action in each cell. This is the same policy evaluated in Homework 3; however, here we assume that the model is unknown to the agent and the setting is episodic. *Notice that for running the model-free algorithms, there should still be an environment model that can generate the samples. For instance, you may have an environment function in your code that takes a state, along with an action, and outputs a next state sampled according to the true (unknown to the agent) transition probabilities and a reward according to the true (unknown to the agent) reward function.*



1. Run a first-visit Monte Carlo algorithm to evaluate this policy. Describe how you ensure that each state is visited often so that its value is estimated accurately. Pick a termination criteria and justify it. Report the value function at all states after the algorithm terminates.
2. Run a one-step temporal-difference learning algorithm to evaluate this policy. Describe how you ensure that each state is visited often so that its value is estimated accurately. Pick a termination criteria and justify it. Report the value function at all states after the algorithm terminates.
3. Compute the true value function and report it at all states.
[Hint: You can apply the analytical solution for policy evaluation but it requires some modifications since this is an episodic setting and not an infinite-horizon setting. To view the problem as an infinite-horizon one and apply the analytical solution, you may consider an extra (fictitious) state where any actions in the terminal states takes the state to this fictitious one. All actions in this new state will keep the agent in the same state and have a reward of 0.]
4. Plot the sequence of errors in the value function with respect to the number of episodes for both algorithms (Monte Carlo and temporal-difference learning) in the same figure. In particular, plot $\|V_n - V^\pi\|_2$ against $n \in \mathbb{N}_0$, where V_n is the value function after n episodes and V^π is the true value function computed in Part 3.

[Note: For this problem, attach all your code in a programming language of your choice to the end of your submission. You may get help from or use the existing implementations of the Monte Carlo and the temporal-difference learning algorithms.]